6 (a) $\dfrac{\text{work done or energy (transformed) (from electrical to other forms)}}{\text{charge}}$ B1 [1]

(b) (i) 1. $V = IR$ *or* $E = IR$ C1

$I = 14/6.0$
$= 2.3\ (2.33)\,\text{A}$ A1 [2]

2. total resistance of parallel resistors = 8.0 Ω C1

current $= 14/(6.0 + 8.0)$
$= 1.0\,\text{A}$ A1 [2]

(ii) $P = EI$ (*allow* $P = VI$) *or* $P = V^2/R$ *or* $P = I^2R$ C1

change in power = $(14 \times 2.33) - (14 \times 1.0)$
or $(14^2/6.0) - (14^2/14)$
or $(2.33^2 \times 6.0) - (1.0^2 \times 14)$

= 19 W (18 W if 2.3 A used) A1 [2]

(c) $I = Anvq$

ratio $= (0.50n/n) \times (1.8A/A)$ or ratio $= 0.50 \times 1.8$ C1

$= 0.90$ A1 [2]